

Flight Delay Analysis & Prediction

Executive Summary

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Overview

This project analysed nearly **3 million domestic U.S. flight records** from 2019 to 2023 to understand the patterns behind flight delays and build predictive tools that can estimate whether – and by how much – a flight will be delayed. The work covers data cleaning, pattern discovery, statistical testing, and the development of five families of machine learning models, all using pre-flight scheduling information only.

Goal: Give airlines and airports actionable insight into when and where delays are most likely, and provide models that can flag high-risk flights before departure.

Key Findings

When delays happen

- Delays build up throughout the day – **early morning flights are the most punctual**; late evening flights are the worst affected.
- **Summer (June–July)** is the worst season; autumn (September–November) is the best.
- A secondary peak appears in **December**, driven by holiday travel.
- **Fridays and Sundays** have the highest delay rates; Tuesdays and Wednesdays the lowest.

Where delays happen

- Some routes exceed a **55% delay rate**, indicating chronic congestion on specific corridors.
- High-traffic airports are **not necessarily the most delayed** – operational efficiency matters more than volume.
- Delay patterns differ significantly across airlines, suggesting carrier-specific operational practices play a role.

Statistically confirmed

Two patterns were formally tested and confirmed with very high certainty (greater than 99.9% confidence):

- **Weekends vs. weekdays:** Weekend flights average 30 seconds more delay than weekdays. While the difference sounds small, with millions of flights it represents a consistent, systematic pattern – not random variation.
 - **Post-pandemic impact:** Flights in 2022–2023 are delayed by **approximately 2 minutes more on average** than pre-pandemic 2019 flights. The aviation industry has not fully recovered to its pre-pandemic punctuality levels.
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Predictive Models

Five types of machine learning models were built and compared. Each was asked to do two things: (1) **estimate the number of minutes of delay** (regression) and (2) **classify the flight** as on-time, short delay, or long delay (classification).

Model	Description	Delay class accuracy	Delay estimate error
Random Forest	Many decision trees voting together	56.8%	29.8 min
LightGBM	Fast, sequential tree learning	54.5%	29.8 min
Neural Network	Multi-layer learning model (deep learning)	54.1%	30.0 min
Decision Tree	Single interpretable tree (baseline)	51.0%	30.1 min
k-Nearest Neighbours	Pattern matching on similar past flights	41.4%	32.2 min

Table 1: Model comparison. “Delay estimate error” is the average error in minutes on the test set.

Random Forest is the best at classifying delay severity; **LightGBM** is equally strong and trains significantly faster. Both are recommended for deployment.

What the accuracy numbers mean

A random guess among three delay categories would be right **33% of the time**. Our best model reaches **56.8%** – nearly twice as good as chance. The average prediction error of **≈30 minutes** reflects an important limitation: using only scheduled timetable data, the models do not know about real-time events such as weather, crew issues, or upstream delays. Adding live data feeds would substantially improve accuracy.

Recommendations

1. **Schedule buffer time strategically.** Build extra turnaround time into schedules for **late-evening departures in summer** and on **Friday/Sunday high-traffic routes**, where delay risk is highest.
2. **Prioritise recovery operations in autumn.** September–November consistently shows the lowest delays. This is the optimal window for maintenance, crew training, and schedule restructuring to restore buffers depleted by the summer peak.
3. **Focus on high-delay routes.** Several routes exceed a 55% delay rate. Targeted operational audits of these corridors – including gate allocation, crew scheduling, and aircraft turnaround – would yield the greatest impact.

Limitations & Next Steps

The models rely solely on pre-flight scheduling information. The primary causes of severe delays – weather events, mechanical issues, crew shortages, and upstream propagation – are not captured in the current feature set. As a result, the models are best used as **risk indicators** rather than precise minute-by-minute forecasts.

Future work could incorporate real-time departure delay signals, weather data, and airport capacity metrics to build a second-generation model with materially higher accuracy.

Full technical report, source code, and trained models are available in the project repository. Dataset: U.S. Department of Transportation, Bureau of Transportation Statistics (via Kaggle), 2019–2023.